FOUR CHARACTERIZATIONS OF COHERENT MODULES ON LOCALLY NOETHERIAN STACKS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Compare quasi-coherent restrictions. Sheaves on Stacks, Lemma stacks-sheaves-lemma-compare-quasi-coherent. x^*F as a pullback of the U-restriction.
- Coherent modules on locally Noetherian algebraic spaces. Cohomology of Spaces, Lemma spaces-cohomology-lemma-coherent-Noetherian. Supports (1)=>(3).
- Finite presentation vs finite type on ringed topoi. Modules on Sites lemmas; definition. (2)=>(1).

Source reference. Original reference 06R6: sources/stacks-chow/tags.txt, source lines 8630–8630. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0DQJ: sources/stacks-chow/tags.txt, source lines 17189–17189. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0GQD: sources/stacks-chow/tags.txt, source lines 20858–20858. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03DO: sources/stacks-chow/tags.txt, source lines 4427–4427. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 07UB: sources/stacks-chow/tags.txt, source lines 9966–9966. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{X}$ be a locally Noetherian algebraic stack. Let $\mathcal{F}$ be an $\mathcal{O}_\mathcal{X}$-module. The following are equivalent*

(1) *$\mathcal{F}$ is a quasi-coherent, finite type $\mathcal{O}_\mathcal{X}$-module,*
(2) *$\mathcal{F}$ is an $\mathcal{O}_\mathcal{X}$-module of finite presentation,*
(3) *$\mathcal{F}$ is quasi-coherent and for any morphism $f : U \to \mathcal{X}$ where U is a locally Noetherian algebraic space, the pullback $f^*\mathcal{F}|_{U_{étale}}$ is coherent, and*
(4) *$\mathcal{F}$ is quasi-coherent and there exists an algebraic space U and a morphism $f : U \to \mathcal{X}$ which is locally of finite type, flat, and surjective, such that the pullback $f^*\mathcal{F}|_{U_{étale}}$ is coherent.*

Date: stacks-project:stacks-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Let $f : U \to \mathcal{X}$ be as in (4). Then U is locally Noetherian (Morphisms of Stacks, Lemma 06R6) and we see that the statement of the lemma makes sense. Additionally, f is locally of finite presentation by Morphisms of Stacks, Lemma 0DQJ. Let x be an object of $\mathcal{X}$ lying over the scheme V. In order to prove (2) we have to show that, after replacing V by the members of an fppf covering of V, the restriction $x^*\mathcal{F}$ has a global finite presentation on $\mathcal{X}/x \cong (Sch/V)_{fppf}$. The projection $W = U \times_{\mathcal{X}} V \to V$ is locally of finite presentation, flat, and surjective. Hence we may replace V by the members of an étale covering of W by schemes and assume we have a morphism $h : V \to U$ with $f \circ h = x$. Since $\mathcal{F}$ is quasi-coherent, we see that the restriction $x^*\mathcal{F}$ is the pullback of $h^*_{small}(f^*\mathcal{F})|_{U_{étale}}$ by π_V, see Sheaves on Stacks, Lemma 0GQD. Since $f^*\mathcal{F}|_{U_{étale}}$ locally in the étale topology has a finite presentation by assumption, we conclude (4) $\Rightarrow$ (2).

Part (2) implies (1) for any ringed topos (immediate from the definition). The properties "finite type" and "quasi-coherent" are preserved under pullback by any morphism of ringed topoi, see Modules on Sites, Lemma 03DO. Hence (1) implies (3), see Cohomology of Spaces, Lemma 07UB. Finally, (3) trivially implies (4). $\square$